Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	61536003-0ad1-472a-bf20-360659f597b8
Date	2026-11-01 18:03:36
Status	Reviewed by expert (expert)

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Tool Wear and Die Damage

Reason: The presence of scratches (32.0 mm2) and major dents (64.0 mm2), together with dimensional deviations on edge L2 (-0.114 mm) and hole H1 (-0.054 mm), directly corresponds to evidence that tool wear and die damage cause surface scratches, dents, and dimensional variations [seed-003].

Effect: Scratches, Dents, and Dimensional Variations

Confidence: 68%

Supporting evidence
- **seed-003**: Tool Wear and Die Damage cause scratches, dents and dimensional variations. Preventive actions: regular tooling inspection, preventive maintenance schedule, die regrinding.

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake
2. Verify material lot

Preventive actions (stop it recurring)

1. Schedule monthly calibration
2. Incoming inspection

Precautions

- Inspect tool and die condition prior to initiating production runs

How to verify the fix

Re-inspect surface finish and verify dimensional compliance of feature L2 and H1 post-regrinding.

4. Other causes considered

Rank	Cause	Confidence	Reason

2	Material Inconsistency and Hardness Variation	65%	The critical crack (56.0 mm2) and major deformation (400.0 mm2) detected on the SPCC steel bracket match the evidence indicating that material inconsistency and hardness variation lead to unpredictable deformation and cracking [seed-002].
3	Improper Bending Force or Over Bending	62%	Major deformation (400.0 mm2) and out-of-tolerance dimensions on edge L2 (-0.114 mm deviation) and hole H1 (-0.054 mm deviation) correspond to evidence stating that improper bending force or over bending causes deformation and dimensional inaccuracy [seed-001].

5. Traceability

AI model: gemini:gemini-3.5-flash. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.